
Subliminal Learning Attacks on Zero-Knowledge Self Proving Models

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Abstract

Self-proving language models train an LLM as the prover in an interactive proof, giving per-input correctness guarantees that distributional accuracy bounds cannot. We consider a natural continuation of this in zero-knowledge proofs, where the prover demonstrates a claim without revealing the witness — a paradigm that will become even more pertinent as LLMs are deployed in settings that require both verifiability and selective disclosure. Zero-knowledge protocols often rely on the independence of the prover’s query distribution from the witness. However, trained LLMs do not always produce true randomness: their output distributions may carry structured deviations that can leak the very information the protocol was designed to protect. Our work aims to study LLM provers in the context of the GMW Graph Isomorphism Protocol, and illustrate that a simple coordinate-wise extractor, utilizing marginal estimation and the Hungarian algorithm, is able to recover the witness from a naively trained prover at rates orders of magnitude above random guessing. We attribute this leakage to the underlying spectral structure of trained transformer log-probability matrices, and prove that under the empirical power-law spectral decay of LLMs, the trained prover’s response distribution is far from uniform. Finally, we propose two defenses, simulator-aligned training and witness masking, that close the leak in our experiments without compromising the prover’s ability to convince an honest verifier.

1 Introduction

Large language models (LLMs) are increasingly being utilized for downstream tasks and applications, where either agents or other modules act upon their outputs. These systems rely on the correctness of LLMs, making it highly advantageous for the outputs to be provably correct, or verifiable. Distributional accuracy bounds do not provide guarantees for a particular input and output pair, motivating recent work to produce checkable answers. An area of focus driven by this idea is production through formal languages whose correctness can be verified through a proof checker [13, 18]. However, these models, given their underlying formal languages, only work for a specific proof system and a specific checker, which is different from the more general case of arbitrary model correctness. This can be achieved through prover-verifier games [3]. Amit et al. [2] expand on this direction via Self-Proving models, which are trained with the goal of convincing a predefined verifier V of the correctness of their outputs through an interactive proof. The soundness of V differs from distributional accuracy and gives per-input correctness guarantees.

In addition to the soundness guarantees provided by interactive proofs, they can also be utilized for privacy. The classical theory of interactive proofs includes zero-knowledge proofs [9], in which the prover aims to convince the verifier of the truthfulness of a statement without revealing any unnecessary information. The privacy gap becomes increasingly crucial as LLM capabilities continue to increase, thus making it even more important that these LLMs are able to provably make claims without giving away any details on how to exploit those claims. An example of this is Anthropic’s recent Claude Mythos Preview [4], which autonomously discovers software vulnerabilities, and was unable to be deployed to the general public to avoid unwanted software exploits. Similar to previous work [2], we do not focus on which scenarios can be modeled through a zero-knowledge interactive proof, but instead focus on the data vulnerabilities of these models and ultimately methods to overcome them.

Zero-knowledge interactive protocols for a given problem are designed on the basis that a polynomial-time simulator, given only the initial problem, can output a sequence that is computationally indistinguishable from the interaction between the prover and the verifier. Computational indistinguishability in these scenarios often relies on queries being independent of both the problem and any intermediate reasoning, especially in cases where the queries are intended to be derived from true randomness. However, true randomness is often not achieved by naively trained LLMs, and attacks have been shown to take advantage of the known specific structure of LLM outputs, allowing for the extraction of proprietary information even from large, widely-used LLMs [6, 8].

This paper exploits the “randomness” used by LLMs in interactive proof protocols to attack zero-knowledge interactions. We make the following contributions: (1) construction of witness extractors against a transformer trained as a GMW-GI prover that are able to recover the secret isomorphism with a probability significantly higher than random guessing, (2) proof that under the empirical power-law spectral decay of LLMs [11], the trained prover’s log-probability matrix admits a quantitative deviation from the honest baseline, and (3) a proposal of two defenses to truly ensure zero-knowledge — simulator-aligned training, which trains the prover against a uniform reference distribution, and witness-masking, ensuring that the LLM doesn’t attend to either the input nor the witness when generating randomness.

2 Related Work

Interactive proofs and zero-knowledge. Interactive proofs [10] consist of two parties: a computationally powerful prover, P , and a computationally efficient verifier V , operating in polynomial time. Given an instance $x \in \mathcal{X}$ of some problem setting, the prover seeks to convince the verifier of the correctness of a claimed solution to x through a protocol consisting of R rounds of interaction. For problems in NP, such proofs could be trivially constructed in just a single round if the prover were to directly transmit the witness to the verifier.

Zero-knowledge interactive proofs extend this framework by introducing an additional requirement: the prover must convince the verifier without revealing any additional information beyond the validity of the statement being proven. For NP languages, this precludes transmission of the witness directly. Formally, zero knowledge is defined by the existence of a polynomial time simulator which, given the problem instance x , generates transcripts which are computationally indistinguishable from transcripts coming from interactions between a prover P and an honest verifier V . It has already been established that zero-knowledge interactive proof protocols exist for every problem in NP [9].

Prover-Verifier LLMs. Prover-verifier frameworks have recently begun to be utilized within large language model contexts in two primary ways. The first adaptation of these frameworks includes employing prover-verifier interactions to improve distributional guarantees during training, with the goal of improving model reliability and interpretability in expectation over a task distribution [14–16]. The second focuses on the production of correctness guarantees given individual problem instances [2, 3]. Such instance-level guarantees are particularly crucial as AI applications begin to be deployed in high-stakes domains such as healthcare, autonomous driving, and legal reasoning, where correctness on individual problem instances is essential.

Amit et al. [2] train an LLM to act as a prover relative to a predefined verifier, ensuring that the model is capable of producing valid responses over R rounds of an interactive proof protocol. Their analysis considers the setting of proving the greatest common divisor of two integers, a situation that requires only a single round of interaction and admits a deterministic protocol. We alternatively explore a more general setting in which soundness demands both multiple rounds of interaction and the prover to incorporate randomness throughout the protocol.

Subliminal learning and low-rank structure in LLMs. Limitations in the expressibility of language models have been studied even prior to the emergence of modern LLMs. Yang et al. [19] demonstrated that the expressiveness of any Softmax-based model, including RNNs and transformers, is constrained by the dimensionality of its word embedding space. Subsequent work exploited these constraints to uncover proprietary information, such as hidden embedding dimensionality [8] or even the embedding projection matrix itself [6].

Other work has identified a low-rank structure not only at the token level, but also in the probability matrix of entire sequences produced by an LLM. Golowich et al. [11] demonstrated that matrices of log-probabilities over prompts and candidate continuations exhibit power-law spectral decay across a wide range of modern LLMs, with exponents slightly exceeding one half. This idea was leveraged to develop a framework [1] to explain the phenomenon of subliminal learning observed by Cloud et al. [7], where a student language model, that is fine-tuned on outputs generated by a teacher model that harbors a hidden trait, inherits that hidden trait, even when the fine-tuning data bears no explicit semantic relation to the hidden trait.

3 Attacking Zero-Knowledge

In this section, we formally define zero-knowledge, outline the requirements for an attack to successfully violate zero-knowledge, and conclude with an outline of potential defensive strategies. We also supply intuition for why such non-random structure may arise in practice.

3.1 Zero-Knowledge for an LLM Prover

Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be an NP relation, where $\mathcal{X}$ is the instance space and $\mathcal{W}$ is the witness space. For any instance $x \in \mathcal{X}$, we define $\mathcal{R}(x) = \{w \in \mathcal{W} : (x, w) \in \mathcal{R}\}$ to be the set of witnesses for x . We consider an interactive proof for $\mathcal{R}$ between a prover and a verifier. Our analysis assumes completeness and soundness throughout and focuses specifically on the zero-knowledge property.

The prover, denoted P , is modeled as an LLM represented by a conditional distribution over transcript continuations. Given a transcript prefix, the prover P samples the next continuation from this distribution. Following Goldwasser et al. [10], we define zero-knowledge through the existence of an efficient simulator without access to a witness that is capable of producing transcripts that are computationally indistinguishable from genuine interactions.

Definition 3.1 (Zero-knowledge for an LLM prover). Let P be the prover and V^* a (possibly malicious) probabilistic polynomial-time (PPT) verifier. Let $\text{View}_{V^*}^P(x)$ denote the random variable that corresponds to V^* 's view of an interaction with P on common input x . We say P is *zero-knowledge* for the interactive proof if there exists a PPT simulator Sim such that for every PPT distinguisher $\mathcal{D}$ with oracle access to P and every $x \in \mathcal{X}$,

$$|\Pr[\mathcal{D}(x, T_1, \dots, T_N) = 1] - \Pr[\mathcal{D}(x, \tilde{T}_1, \dots, \tilde{T}_N) = 1]| \leq \text{negl}(n), \quad (1)$$

where $T_k \stackrel{\text{iid}}{\sim} \text{View}_{V^*}^P(x)$ and $\tilde{T}_k \stackrel{\text{iid}}{\sim} \text{Sim}(x)$.

Assumption 3.2 (Black-box access to the prover). The malicious verifier has oracle access to P and can send any query x . Given a query $x \in \mathcal{X}$, the model internally computes a witness φ via annotated transcript learning [2], after which the oracle returns samples from $P[\cdot \mid x, \varphi]$.

3.2 Graph Isomorphism (GI)

Graph Isomorphism (GI) provides a structural property that aids in our analysis: the witness and the commitment randomness are permutations in S_n , and under the cheating-verifier branch, the commitment randomness is revealed verbatim.

The instance consists of a pair of graphs $x = (G_0, G_1)$ defined over vertex set $[n]$. The witness is the permutation $\varphi \in S_n$ satisfying $\varphi(G_0) = G_1$. Recovering φ from x is precisely the graph isomorphism problem. The protocol of Goldreich et al. [9], restricted to the challenge $c = 0$, proceeds as follows:

1. The prover samples $\psi \stackrel{\text{iid}}{\sim} S_n$.
2. The verifier sends challenge $c = 0$.
3. The prover reveals $\sigma = \psi$ such that $H = \sigma(G_0)$.

On input x , the LLM generates the transcript token-by-token in three stages per round:

1. Internally compute a witness $\varphi \in S_n$ via chain-of-thought reasoning (the annotation learning step of [2] which is importantly not exposed to the verifier).
2. Sample commitment randomness $\psi \sim P[\cdot \mid x, \varphi]$.
3. Complete the verifier's challenge by emitting the appropriate response tokens, which is ψ if $c = 0$.

3.3 Witness Extraction Breaks Zero-Knowledge

We now formalize the connection between extraction and zero-knowledge: any polynomial-time algorithm that is able to recover the witness from polynomially many transcripts with non-negligible probability contradicts zero-knowledge, provided the underlying relation is average-case hard.

Assumption 3.3 (Average-case hardness of $\mathcal{R}$). There exists an instance distribution $\mathcal{I}$ over $\mathcal{X}$ such that for every PPT algorithm $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{W}$,

$$\Pr_{(x, \varphi) \sim \mathcal{I}} [\mathcal{A}(x) \in \mathcal{R}(x)] \leq \text{negl}(n). \quad (2)$$

For GI, average-case hardness remains conjectural and lacks a known canonical hard distribution [5]. However, similar permutation-based protocols exhibit conjectural average-case hardness with well-studied hard distributions, such as PKP, for which Theorem 3.3 is the basis of the standard PKP hardness assumption. For simplicity, we take this assumption to hold for some distribution $\mathcal{I}$ for the remainder of the paper.

Lemma 3.4 (Extraction breaks zero-knowledge). *Suppose Theorems 3.2 and 3.3 hold. Assume there exists a PPT algorithm EXTRACT, with oracle access to P , such that for some $N = \text{poly}(n)$,*

$$\Pr_{\substack{(x, \varphi) \sim \mathcal{I} \\ \psi_1, \dots, \psi_N \sim P[\cdot \mid x']}} [\text{EXTRACT}(x, \psi_1, \dots, \psi_N) \in \mathcal{R}(x)] > \text{negl}(n) \quad (3)$$

where x' can be any query to the LLM. Then P is not zero-knowledge.

The full proof, deferred to Section A, follows the standard simulator-to-witness-finder reduction: zero-knowledge transfers EXTRACT's success from real transcripts to simulated transcripts, and running EXTRACT on simulator transcripts yields a PPT witness finder which contradicts Theorem 3.3. This reduces the problem of attacking zero-knowledge to the challenge of constructing a successful EXTRACT. The next subsection gives one such construction and identifies the precise property of P under which it works.

Algorithm 1 $\text{EXTRACT}(x, \psi_1, \dots, \psi_N)$ with advice $\{(i_j, v_j)\}_{j \in [n]}$

Require: Instance x ; coordinates $\{(i_j, v_j)\}$; oracle access to P .

- 1: **for** $j \in [n]$ **do**
 - 2: Estimate $\hat{\tau}_j$ for τ_{i_j, v_j} (Equation (5)) and $\hat{p}_j$ for p_{i_j, v_j}^* (Equation (4)) via oracle queries.
 - 3: Set $\hat{L}_j \leftarrow \log(\hat{p}_j)$.
 - 4: **end for**
 - 5: Build $C[j, u] \leftarrow |\hat{L}_j - \hat{\tau}_j(u)|$.
 - 6: **return** $\hat{\varphi} \leftarrow \arg \min_{\sigma \in S_n} \sum_j C[j, \sigma(j)]$ via the Hungarian algorithm.
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3.4 A Coordinate-Wise Extractor

If $P[\psi \mid x, \varphi]$ was exactly uniform on S_n , no information about φ leaks and no extraction is possible. The attack therefore relies on witness-correlated deviations from the uniformity in the LLM’s distribution, which is a phenomenon we observe empirically in trained models (Section 5).

For $i, v \in [n]$ and $(x, \varphi) \in \mathcal{R}$, let

$$p_{i,v}^*(x, \varphi) = \Pr_{\psi \sim P[\cdot \mid x, \varphi]}[\psi(i) = v] \quad (4)$$

denote the prover’s marginal probability of placing value v at coordinate i . For indices i, j, v , let

$$\tau_{i,j,v}(u) = \mathbb{E}_{\varphi' \sim \text{Unif}(S_n \mid \varphi'(j)=u)}[\log p_{i,v}^*(x_{\varphi'}, \varphi')], \quad (5)$$

where $x_{\varphi'}$ is the instance induced by witness φ' (e.g., $(G_0, \varphi'(G_0))$ for GI). This table can be estimated from oracle queries to P via Monte Carlo as shown in Theorem B.5.

Assumption 3.5 (Log assignment margin). For a non-negligible fraction of $(x, \varphi) \sim \mathcal{I}$, there exist witnessing coordinates $\{(i_j, v_j)\}_{j \in [n]}$ such that the population log-cost matrix

$$C^*[j, u] = |\log p_{i_j, v_j}^*(x, \varphi) - \tau_{i_j, v_j}(u)| \quad (6)$$

satisfies

$$\min_{\sigma \in S_n, \sigma \neq \varphi} \sum_{j=1}^n C^*[j, \sigma(j)] - \sum_{j=1}^n C^*[j, \varphi(j)] \geq \alpha_{\log} \quad (7)$$

for some $\alpha_{\log} > \text{negl}(n)$.

Intuitively, Theorem 3.5 states that the prover’s coordinate marginals encode the true witness φ in the unique-optimal-assignment sense: there is polynomial-size advice (the witnessing coordinates) under which φ is identifiable as the assignment of minimum log-cost, separated by a polynomial margin from every other permutation.

Theorem 3.6 (Extraction under Theorem 3.5). *Suppose Theorems 3.2, 3.3 and 3.5 hold given a polynomial-time advice finder for $\{(i_j, v_j)\}_{j \in [n]}$ (Theorem B.2, appendix). Then Algorithm 1 satisfies the hypothesis of Theorem 3.4: it recovers φ on a non-negligible fraction of instances using $\text{poly}(n)$ oracle queries to P . In combination with Theorem 3.4, P is therefore not zero-knowledge.*

The full proof, including concentration bounds for $\hat{\tau}_j$ and $\hat{L}_j$, is in Section B.

3.5 Why Structure Exists: Subliminal Learning

Theorem 3.5 requires the prover’s response distribution to encode the witness in a coordinate-aligned way. We now argue that such structural deviation from uniform sampling arises naturally and is generic for trained LLM provers, drawing on recent results on subliminal learning [1, 7] and the spectral structure of transformer log-probability matrices [11].

For an honest GMW-GI prover that samples $\psi \sim \text{Unif}(S_n)$ independently of (x, φ) , the log-probability matrix

$$X_M[\psi, (x, \varphi)] = \log \Pr_M[\psi \mid x, \varphi] \quad (8)$$

indexed by $\psi \in S_n$ (rows) and $(x, \varphi) \in \mathcal{V} = \{(x, \varphi) : \varphi \in \mathcal{R}(x)\}$ (columns) is the constant matrix with all entries equal to $-\log(n!)$. This matrix has rank one, and this rank-one structure precisely captures zero-knowledge under challenge $c = 0$ since the distribution depends neither on ψ nor on (x, φ) .

Golowich et al. [11] measure the singular value spectrum of log-probability matrices for diverse trained transformers and report that the normalized singular values follow a power-law decay $\sigma_k/\sigma_1 \approx k^{-\alpha}$ with exponent $\alpha > 1/2$. Building on this, Aden-Ali et al. [1] formalize subliminal learning through low-rank structural properties of matrices.

Two consequences follow from this spectral assumption. The full statements (as well as statements for sub-matrices of X_M) and proofs are in Section C and we summarize the conclusions here.

Theorem 3.7 (Frobenius spectral deviation; informal). *Under power-law spectral decay with exponent $\alpha > 1/2$, the trained prover’s log-probability matrix differs from the honest baseline in Frobenius norm by at least*

$$\|X_M - X_M^{\text{honest}}\|_F = \Omega\left(\log(n!) \cdot \sqrt{|S_n| \cdot |\mathcal{V}|}\right) \quad (9)$$

Corollary 3.8 (Average χ^2 deviation from uniform; informal). *Under the same assumption together with a bounded-probability regularity condition,*

$$\mathbb{E}_{(x,\varphi) \sim \text{Unif}(\mathcal{V})} [\chi^2(p_{x,\varphi}, \text{Unif}(S_n))] \geq \Omega((\log n!)^2) = \text{poly}(n). \quad (10)$$

where $p_{x,\varphi}$ represents the probability distribution sampled by $\text{Pr}_M[\psi \mid x, \varphi]$

The trained prover’s response distribution is therefore far from uniform on average, with the deviation polynomial in n . We note that Theorem 3.8 is a divergence statement, not a sample complexity bound. It does not alone imply efficient, namely polynomial-time, extraction of the witness but rather that the structural deviation is real and quantitatively large. We empirically test the remaining claim that there exists some polynomial-time detectable component of this deviation that aligns with the witness in the sense of Theorem 3.5.

The conjecture is also motivated by literature on subliminal learning. Cloud et al. [7] demonstrate that latent traits of a teacher model (e.g., a preference for owls) can be transferred to student models when fine-tuned on the teacher’s outputs – even when those outputs are seemingly unrelated to the trait, such as random number sequences. Aden-Ali et al. [1] further formalize this phenomenon and demonstrate how to systematically construct datasets capable of eliciting hidden behaviors via fine-tuning. In both cases, fine-tuning is the polynomial-time procedure that extracts the latent information from the teacher’s output distribution.

4 Defending Zero-Knowledge

We now formalize two defenses against the witness-extraction attack introduced in Section 3.4.

4.1 Simulator-aligned training

The honest GMW-GI prover samples commitment randomness via $\psi \sim \text{Unif}(S_n)$. For the tokens generating ψ , we replace the standard cross-entropy objective with a KL divergence loss against the autoregressive factorization of $\text{Unif}(S_n)$. Concretely, the first coordinate is sampled uniformly from $[n]$, the second is uniformly from $[n] \setminus \{\psi(0)\}$, and each subsequent coordinate is sampled uniformly from the remaining unassigned elements. All remaining tokens are trained as outlined in Amit et al. [2].

This defense requires no architectural modifications; however, it’s reliant upon a coordinate-aligned tokenization scheme such that each coordinate of ψ is represented by a single token and the autoregressive uniform target remains explicitly expressible.

4.2 Witness masking

A complementary defense modifies the model’s attention mask during generation of ψ such that those tokens cannot attend to the internal chain-of-thought witness or to G_1 (which carries information about φ via $G_1 = \varphi(G_0)$). The mask is then removed for subsequent generation, since responding to challenge $c = 1$ requires both φ and ψ . Under this masking strategy, $P[\psi \mid x, \varphi] = P[\psi \mid G_0]$, so $\tau_{i,j,v}(u)$ is constant in u , eliminating the coordinate-dependent structure required by Theorem 3.5. As a result, the extraction attack on Section 3.4 can no longer succeed.

Unlike simulator-aligned training, this defense is agnostic to tokenization schemes and is able to be applied more broadly. However, that tradeoff is that it requires modification of the model’s attention mechanism at both training and inference time.

5 Experiments

We empirically demonstrate witness extraction against an LLM prover trained as a GMW-GI prover, evaluating multiple extraction algorithms across various witness sizes and defense conditions. We compare three training regimes: a baseline prover trained with a standard loss, a simulator-aligned prover (Section 4.1), and a witness-masked prover (Section 4.2).

5.1 Experimental Setup

We train a small transformer (TinyTransformer, see Section D) as a prover for the GMW-GI protocol on graph sizes $4 \leq n \leq 9$. Following Amit et al. [2], the model produces a witness through an internal chain-of-thought process before

sampling the commitment randomness ψ . To make evaluation tractable at larger values of n , we train the model to predict ψ conditioned on the witness φ alone, omitting the graphs G_0 and G_1 from the context. For smaller graph sizes n (verified for $n \leq 6$), we found that this simplification did not affect witness recovery rates relative to the full conditioning on (G_0, G_1, φ) . Without this simplification, evaluating at $n = 7$ and beyond would require multiple weeks of computations.

In addition, learning to solve for the permutation from G_0 to G_1 is often a very difficult problem. We found that for $n = 6$, the accuracy percentage for a correct permutation was only 30%, so to focus on the zero-knowledge aspect, we gave the model φ as an input. This also prevented the confounding factor of multiple possible witnesses. More details can be found at Section D.

We evaluate six extractors constructed from the same collection of τ tables, varying along two dimensions: the method used to construct the cost matrix and whether the construction operates on raw probabilities or log probabilities. Let $p(i, v) = \Pr_{\psi \sim P[\cdot|x, \varphi]}[\psi(i) = v]$ and $L(i, v) = \log(p(i, v))$.

- **Single-witness, max-spread (Algorithm 1).** For each j , we select witnessing coordinates (i_j, v_j) maximizing $\max_u \tau_{i_j, j, v_j}(u) - \min_u \tau_{i_j, j, v_j}(u)$. The resulting cost matrix is defined as $C[j, u] = |L(i_j, v_j) - \tau_{i_j, j, v_j}(u)|$ (log) or $|p(i_j, v_j) - \tau_{i_j, j, v_j}(u)|$ (raw where τ is computed without the log).
- **Aggregate- L^p .** Rather than selecting a single witnessing coordinate pair, we incorporate all (i, v) pairs jointly: $C[j, u] = (\sum_{i, v} |\hat{L}(i, v) - \tau_{i, j, v}(u)|^p)^{1/p}$ for $p \in \{1, 2\}$, for log-probability; raw-probability is defined similarly for the non-logged probabilities.

Together, these constructions yield six extractors. All methods can recover either the best estimated witness $\hat{\varphi}$ by utilizing the Hungarian algorithm or the set of top- n witnesses by utilizing Murty’s algorithm [17] on the resulting $n \times n$ cost matrix.

5.2 Results

Table 1 reports witness-recovery rates for the baseline prover for $n \in \{4, \dots, 9\}$. Random-guessing baselines are $n/n! = 1/(n-1)!$ for top- n extraction. The extractor succeeds significantly above the random baseline across all witness sizes and extraction methods, demonstrating that the witness-correlated structure required by Theorem 3.5 is present in naively trained provers. The decrease in the percentage of improvement relative to the random baseline from $n = 7$ to $n = 8$ is likely due to the training dataset size as described in Section D.4.

Table 1: Top- n witness recovery rate (% , multiplier relative to random baseline) on the baseline prover. Random baseline is $100/(n-1)!$.

Method	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$
Random baseline (%)	16.67	4.17	0.83	0.14	1.98×10^{-2}	2.48×10^{-3}
Single, max-spread (raw)	45.83 (2.8 $\times$)	7.50 (1.8 $\times$)	2.22 (2.7 $\times$)	0.44 (3.1 $\times$)	0.04 (1.8 $\times$)	0.05 (20 $\times$)
Single, max-spread (log)	66.67 (4.0 $\times$)	17.50 (4.2 $\times$)	3.75 (4.5 $\times$)	0.79 (5.7 $\times$)	0.03 (1.5 $\times$)	0.05 (20 $\times$)
Agg L^1 (raw)	87.50 (5.2 $\times$)	51.67 (12 $\times$)	17.64 (21 $\times$)	6.31 (45 $\times$)	0.31 (16 $\times$)	0.10 (40 $\times$)
Agg L^1 (log)	79.17 (4.8 $\times$)	43.33 (10 $\times$)	15.42 (19 $\times$)	5.74 (41 $\times$)	0.08 (3.8 $\times$)	0.15 (60 $\times$)
Agg L^2 (raw)	95.83 (5.8 $\times$)	49.17 (12 $\times$)	17.64 (21 $\times$)	10.10 (73 $\times$)	0.31 (15 $\times$)	0.20 (80 $\times$)
Agg L^2 (log)	95.83 (5.8 $\times$)	49.17 (12 $\times$)	17.64 (21 $\times$)	10.10 (73 $\times$)	0.17 (8.6 $\times$)	0.15 (60 $\times$)

Table 2 reports recovery rates for the same extractors against the two defended provers. Both defenses reduce recovery to the random baseline across all methods and model sizes tested.

Table 2: Top- n witness recovery rate (%) on the defended provers. Random baselines match Table 1.

Method	Simulator-aligned			Witness-masked		
	$n = 4$	$n = 5$	$n = 6$	$n = 4$	$n = 5$	$n = 6$
Random baseline	1.67×10^1	4.17×10^0	8.33×10^{-1}	1.67×10^1	4.17×10^0	8.33×10^{-1}
Single-witness, max-spread (raw)	12.50 (0.8 $\times$)	3.33 (0.8 $\times$)	0.97 (1.2 $\times$)	16.67 (1.0 $\times$)	5.00 (1.2 $\times$)	0.83 (1.0 $\times$)
Single-witness, max-spread (log)	12.50 (0.8 $\times$)	5.83 (1.4 $\times$)	0.56 (0.7 $\times$)	25.00 (1.5 $\times$)	4.17 (1.0 $\times$)	0.69 (0.8 $\times$)
Aggregate L^1 (raw)	20.83 (1.2 $\times$)	5.00 (1.2 $\times$)	0.97 (1.2 $\times$)	20.83 (1.2 $\times$)	2.50 (0.6 $\times$)	0.56 (0.7 $\times$)
Aggregate L^1 (log)	16.67 (1.0 $\times$)	2.50 (0.6 $\times$)	1.25 (1.5 $\times$)	16.67 (1.0 $\times$)	1.67 (0.4 $\times$)	0.97 (1.2 $\times$)
Aggregate L^2 (raw)	20.83 (1.2 $\times$)	3.33 (0.8 $\times$)	0.69 (0.8 $\times$)	8.33 (0.5 $\times$)	1.67 (0.4 $\times$)	0.83 (1.0 $\times$)
Aggregate L^2 (log)	20.83 (1.2 $\times$)	2.50 (0.6 $\times$)	0.69 (0.8 $\times$)	20.83 (1.2 $\times$)	3.33 (0.8 $\times$)	0.97 (1.2 $\times$)

5.3 Discussion

The results demonstrate a strong qualitative pattern — standardly trained provers exhibit substantial witness leakage. Specifically, we observe that the recovery rates in Table 1 are significantly above the random baseline for the standardly trained provers consistently across all witness sizes, while the simulator-aligned and witness-masked provers (Table 2) reduce recovery to baseline levels. The attack therefore demonstrates success on naively trained models, but is effectively neutralized by either defense.

These experiments by themselves, however, do not establish a formal violation of zero-knowledge in the asymptotic sense of Theorem 3.1. Two main limitations stand in the way.

First, we trained a separate model for each value of n rather than a single length-generalizing model. In order to observe a closer relation to the asymptotic setting required when showing polynomial success frequencies, we would require the analysis of a length-generalizing prover.

Second, empirically distinguishing polynomial decay from exponential decay would require data across much higher orders of magnitude in the input parameter [12]. Our experiments only cover six values of n (ranging from 4 to 9), which means that we cannot sufficiently and reliably distinguish polynomial decay from exponential decay. Establishing polynomial versus exponential distinction would require substantially larger sizes for n , which are currently compute-bound.

Therefore, we interpret these experiments not as a complete empirical refutation of zero-knowledge, but rather as evidence that the structural assumptions of Section 3.4 are realized in trained provers and that the proposed defenses neutralize the leakage. Strengthening the empirical claim to match the theoretical statement remains an important direction for future work, with the most direct path being a length-generalizing prover trained on a broader range of input sizes.

6 Conclusion

We have shown that LLM provers trained to execute zero-knowledge interactive proofs can leak witness information through the structure of their commitment randomness, that this leakage is recoverable via coordinate-wise extractors based on marginal estimation, and that two natural defenses — simulator-aligned training and witness masking — substantially mitigate this leakage in our experiments. The structural reason for the leak, formalized via the spectral structure of trained transformer log-probability matrices and subliminal learning literature, suggests that this phenomenon is not merely an artifact of our specific setup, but instead a reflection of the generic property of trained LLM provers.

As LLMs become more capable, deployment in settings where they must demonstrate possession of sensitive knowledge without disclosing the knowledge itself is growing rapidly. Examples of these settings may include proving awareness of a software vulnerability, strategic plan, or confidential information, all while withholding the underlying details from the public or any unauthorized group. The cryptographic theory of zero-knowledge offers a toy framework for these settings, yet our results demonstrate that naively learned LLM provers can be exploited. Without explicit defenses adjacent to the ones proposed here, deploying LLM provers in zero-knowledge settings risks silently leaking precisely the information the protocol was designed to protect.

Our results are subject to several important limitations regarding the gap between our finite-sample experiments and the asymptotic statement of zero-knowledge; these limitations are discussed further in Section 5.3. A natural direction for future work is to investigate whether similar leakage occurs in large pretrained frontier models and how this framework could be modified for interactive proof protocols that do not have permutation structures for the witness and randomness.

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A Proof of Theorem 3.4

We restate the theorem for convenience.

Lemma A.1 (Extraction breaks zero-knowledge; restatement of Theorem 3.4). *Suppose Theorems 3.2 and 3.3 hold. Suppose further that there exists a PPT algorithm EXTRACT, with oracle access to P , such that for some $N = \text{poly}(n)$,*

$$\Pr_{\substack{(x, \varphi) \sim \mathcal{I} \\ \psi_1, \dots, \psi_N \sim P[\cdot|x']}} [\text{EXTRACT}(x, \psi_1, \dots, \psi_N) \in \mathcal{R}(x)] > \text{negl}(n) \quad (11)$$

Then P is not zero-knowledge in the sense of Theorem 3.1.

Proof. Suppose for contradiction that P is zero-knowledge with PPT simulator Sim . We derive a PPT algorithm A that contradicts Theorem 3.3.

Let $\mu > \text{negl}(n)$ lower-bound the success probability of EXTRACT on real transcripts as in the theorem hypothesis.

Define the PPT distinguisher

$$\mathcal{D}(x, T_1, \dots, T_N) = \mathbf{1}[\text{EXTRACT}(x, T_1, \dots, T_N) \in \mathcal{R}(x)], \quad (12)$$

where membership in $\mathcal{R}(x)$ is decidable in polynomial time given x and a candidate witness (this is the defining property of an NP relation). Note $\mathcal{D}$ has oracle access to P , which is permitted by Theorem 3.1.

By the theorem hypothesis,

$$\Pr_{(x, \varphi) \sim \mathcal{I}, T_k \sim \text{View}_{V^*}^P(x)}[\mathcal{D}(x, T_1, \dots, T_N) = 1] \geq \mu. \quad (13)$$

By Theorem 3.1 ,

$$|\Pr[\mathcal{D}(x, T_1, \dots, T_N) = 1] - \Pr[\mathcal{D}(x, \tilde{T}_1, \dots, \tilde{T}_N) = 1]| \leq \text{negl}(n), \quad (14)$$

where $T_k \sim \text{View}_{V^*}^P(x)$ and $\tilde{T}_k \sim \text{Sim}(x)$. Averaging over $(x, \varphi) \sim \mathcal{I}$:

$$|\Pr_{\mathcal{I}, P}[\mathcal{D} = 1] - \Pr_{\mathcal{I}, \text{Sim}}[\mathcal{D} = 1]| \leq \text{negl}(n). \quad (15)$$

Hence

$$\Pr_{\mathcal{I}, \text{Sim}}[\mathcal{D} = 1] \geq \mu - \text{negl}(n) > \text{negl}(n) \quad (16)$$

for sufficiently large n .

Define the PPT algorithm $A(x)$:

- (a) Sample $\tilde{T}_k \sim \text{Sim}(x)$ iid for $k \in [N]$.
- (b) Run $\text{EXTRACT}(x, \tilde{T}_1, \dots, \tilde{T}_N)$ and return its output.

A is PPT and takes only x as input. Since EXTRACT outputs an element of $\mathcal{R}(x)$ with non-negligible probability, we have $\Pr_{\mathcal{I}}[A(x) \in \mathcal{R}(x)] = \Pr_{\mathcal{I}, \text{Sim}}[\pi = 1] > \text{negl}(n)$.

Theorem 3.3 states $\Pr_{\mathcal{I}}[A(x) \in \mathcal{R}(x)] \leq \text{negl}(n)$ for every PPT A . The bound above contradicts this. Therefore P is not zero-knowledge. $\square$

B Proof of Theorem 3.6

This appendix gives the full proof of Theorem 3.6, including all concentration bounds. The proof has three main components:

- Concentration of empirical log-marginals around their population values (Theorem B.4).
- Estimation of the conditional log-marginal table $\tau_{i,j,v}$ via Monte Carlo (Theorem B.5).
- Hungarian-optimum stability under cost matrix perturbation, combined with the log assignment margin (Theorem 3.6).

We additionally invoke a polynomial-time advice finder (Theorem B.2), stated below.

B.1 Additional assumptions and notation

We assume the following regularity condition on the prover: a transformer trained to predict a uniform distribution will not output probabilities significantly different from the uniform prediction.

Assumption B.1 (Bounded marginal probability). For every $i, v \in [n]$ and $(x, \varphi) \in \mathcal{R}$:

$$p_{i,v}^*(x, \varphi) \geq 1/(2n). \quad (17)$$

Assumption B.2 (Polynomial-time advice finder). There is a polynomial-time algorithm FINDCOORDS with oracle access to P that, on input x , outputs $\{(i_j, \hat{v}_j)\}_{j \in [n]} \in ([n]^2)^n$ with the following property: whenever (x, φ) satisfies Theorem 3.5 with margin $\alpha_{\log} = 1/\text{poly}(n)$ and witnessing coordinates $\{(i_j, v_j)\}_{j \in [n]}$, then with non-negligible probability over the randomness of FINDCOORDS and oracle samples,

$$\{(i_j, \hat{v}_j)\}_{j \in [n]} = \{(i_j, v_j)\}_{j \in [n]}. \quad (18)$$

The advice $\{(i_j, v_j)\}_{j \in [n]}$ specifies, for each coordinate j , a witnessing observable $\{\psi(i_j) = v_j\}$. We write $\ell_j^* = \log p_{i_j, v_j}^*(x, \varphi)$ for the population log-marginal at the chosen observables.

B.2 Concentration tools

Lemma B.3 (Hoeffding's inequality). *Let $X_1, \dots, X_N$ be iid with $X_k \in [a, b]$ and $\mu = \mathbb{E}[X_k]$. For every $t > 0$,*

$$\Pr \left[\left| \frac{1}{N} \sum_k X_k - \mu \right| \geq t \right] \leq 2 \exp \left(-\frac{2Nt^2}{(b-a)^2} \right). \quad (19)$$

Lemma B.4 (Probability-to-log-probability concentration). *If $\hat{p}$ is the empirical mean of N iid Bernoulli(p^*) samples with $p^* \geq 1/(2n)$, then for any $\eta \in (0, 1/2]$,*

$$\Pr[|\log \hat{p} - \log p^*| \geq 2\eta] \leq 2 \exp \left(-\frac{N\eta^2}{2n^2} \right). \quad (20)$$

Proof. By Theorem B.3 with $t = \eta p^* \geq \eta/(2n)$,

$$\Pr[|\hat{p} - p^*| \geq \eta p^*] \leq \Pr[|\hat{p} - p^*| \geq \eta/(2n)] \quad (21)$$

$$\leq 2 \exp(-N\eta^2/(2n^2)) \quad (22)$$

Assume $|\hat{p} - p^*| < \eta p^*$. Then, $\hat{p} \in [(1-\eta)p^*, (1+\eta)p^*]$, so $\log(1-\eta) \leq \log(\hat{p}/p^*) \leq \log(1+\eta)$. Using $|\log(1 \pm x)| \leq 2|x|$ for $|x| \leq 1/2$, we get $|\log \hat{p} - \log p^*| \leq 2\eta$. Thus, $\Pr[|\log \hat{p} - \log p^*| \geq 2\eta] \leq 2 \exp(-N\eta^2/(2n^2))$. $\square$

B.3 Estimation of the log-marginal table

Lemma B.5 (Estimation of $\tau_{i,j,v}$). *Fix $i, j, v \in [n]$ and an instance x . Under Theorems B.1 and 3.2, there is an algorithm with oracle access to P that, for any $\epsilon, \delta \in (0, 1/2]$, makes $O(n^4 \log^2(n/\delta)/\epsilon^4)$ oracle queries and outputs $\hat{\tau}_{i,j,v} : [n] \rightarrow \mathbb{R}$ with*

$$\Pr \left[\max_{u \in [n]} |\hat{\tau}_{i,j,v}(u) - \tau_{i,j,v}(u)| \geq \epsilon \right] \leq \delta. \quad (23)$$

Proof. Fix $u \in [n]$. The goal is to estimate

$$\tau_{i,j,v}(u) = \mathbb{E}_{\varphi' \sim \text{Unif}(S_n | \varphi'(j)=u)} [\log p^*(\varphi')], \quad (24)$$

where $p^*(\varphi') = p_{i,v}^*(x_{\varphi'}, \varphi')$.

Sample $\varphi'_1, \dots, \varphi'_M$ iid from $\text{Unif}(S_n | \varphi'(j) = u)$ with $M = \lceil 8n^2 \log(4n/\delta)/\epsilon^2 \rceil$. For each m , draw $K = \lceil 32n^2 \log(4nM/\delta)/\epsilon^2 \rceil$ iid samples $\psi_1^{(m)}, \dots, \psi_K^{(m)} \sim P[\cdot | x_{\varphi'_m}, \varphi'_m]$ and set

$$\hat{p}_m = \frac{1}{K} |\{k : \psi_k^{(m)}(i) = v\}|, \quad \hat{\tau}_{i,j,v}(u) = \frac{1}{M} \sum_m \log \hat{p}_m. \quad (25)$$

The estimator's error decomposes into two parts: $\hat{p}_m$ deviates from $p^*(\varphi'_m)$ (inner), and $\frac{1}{M} \sum_m \log p^*(\varphi'_m)$ deviates from its mean $\tau_{i,j,v}(u)$ (outer).

Fix m and condition on φ'_m , so $p^*(\varphi'_m)$ is fixed and $\hat{p}_m$ is the empirical mean of K iid Bernoulli($p^*(\varphi'_m)$) samples. By Theorem B.1, $p^*(\varphi'_m) \geq 1/(2n)$. Apply Theorem B.4 with $\eta = \epsilon/4 \leq 1/2$:

$$\Pr[|\log \hat{p}_m - \log p^*(\varphi'_m)| \geq \epsilon/2 | \varphi'_m] \leq 2 \exp(-K\epsilon^2/(32n^2)) \leq \delta/(2nM), \quad (26)$$

where the last inequality uses $K \geq 32n^2 \log(4nM/\delta)/\epsilon^2$. Taking expectation over φ'_m and union-bounding over $u \in [n]$ and $m \in [M]$,

$$\Pr[\mathcal{E}_{\text{in}}^c] \leq nM \cdot \frac{\delta}{2nM} = \delta/2, \quad (27)$$

where $\mathcal{E}_{\text{in}} = \{|\log \hat{p}_m - \log p^*(\varphi'_m)| \leq \epsilon/2 \text{ for all } u \in [n], m \in [M]\}$.

The variables $\log p^*(\varphi'_m)$ are deterministic functions of the iid φ'_m , hence iid, with common mean $\tau_{i,j,v}(u)$. By Theorem B.1, $\log p^*(\varphi'_m) \in [-\log(2n), 0]$, so the range is $\log(2n) \leq n$ for $n \geq 2$. By Theorem B.3 with $t = \epsilon/2$,

$$\Pr \left[\left| \frac{1}{M} \sum_m \log p^*(\varphi'_m) - \tau_{i,j,v}(u) \right| \geq \epsilon/2 \right] \leq 2 \exp(-M\epsilon^2/(2n^2)) \leq \delta/(2n), \quad (28)$$

using $M \geq 8n^2 \log(4n/\delta)/\epsilon^2$. Union-bounding over $u \in [n]$,

$$\Pr[\mathcal{E}_{\text{out}}^c] \leq n \cdot \frac{\delta}{2n} = \delta/2, \quad (29)$$

where $\mathcal{E}_{\text{out}} = \{|\frac{1}{M} \sum_m \log p^*(\varphi'_m) - \tau_{i,j,v}(u)| \leq \epsilon/2 \text{ for all } u \in [n]\}$.

On $\mathcal{E}_{\text{in}} \cap \mathcal{E}_{\text{out}}$, the triangle inequality gives, for every $u \in [n]$,

$$\begin{aligned} |\hat{\tau}_{i,j,v}(u) - \tau_{i,j,v}(u)| &\leq \frac{1}{M} \sum_m |\log \hat{p}_m - \log p^*(\varphi'_m)| + \left| \frac{1}{M} \sum_m \log p^*(\varphi'_m) - \tau_{i,j,v}(u) \right| \\ &\leq \epsilon/2 + \epsilon/2 = \epsilon. \end{aligned}$$

The total failure probability is $\Pr[\mathcal{E}_{\text{in}}^c \cup \mathcal{E}_{\text{out}}^c] \leq \delta$. The total query count is $MK = O(n^4 \log^2(n/\delta)/\epsilon^4)$. $\square$

B.4 Proof of Theorem 3.6

Theorem B.6 (Restatement of Theorem 3.6). *Suppose Theorems B.1, B.2, 3.2, 3.3 and 3.5 hold. Then Algorithm 1 satisfies the hypothesis of Theorem 3.4: it recovers φ on a non-negligible fraction of instances using $\text{poly}(n)$ oracle queries to P . Combined with Theorem 3.4, P is not zero-knowledge.*

Proof. Let $\mu > \text{negl}(n)$ lower-bound the fraction of $(x, \varphi) \sim \mathcal{I}$ at which Theorem 3.5 holds, with margin $\alpha_{\log} > \text{negl}(n)$. Let $q > \text{negl}(n)$ be the success probability of FINDCOORDS from Theorem B.2. Set $\beta = 1/4$ and allocate $N = \text{poly}(n)$ oracle queries. We show that whenever (a) (x, φ) satisfies Theorem 3.5, (b) FINDCOORDS outputs the correct advice, and (c) the concentration events below hold, Algorithm 1 returns $\hat{\varphi} = \varphi$.

For each coordinate j , apply Theorem B.5 to (i_j, j, v_j) with precision $\epsilon = \alpha_{\log}/(8n)$ and confidence $\delta = \beta/(2n)$. This uses $\text{poly}(n, 1/\alpha_{\log}, \log(1/\beta))$ oracle queries per coordinate, $\text{poly}(n)$ in total. Union-bounding over j , the event

$$\mathcal{E}_\tau = \{|\hat{\tau}_j(u) - \tau_{i_j,j,v_j}(u)| \leq \alpha_{\log}/(8n) \text{ for all } j, u \in [n]\} \quad (30)$$

holds with probability $\geq 1 - \beta/2$ conditional on the correct advice.

For each j , apply Theorem B.4 to the empirical log-marginal $\hat{L}_j$ with $\eta = \alpha_{\log}/(16n)$. Setting $N \geq 512 n^4 \log(4n/\beta)/\alpha_{\log}^2$ gives

$$\Pr[|\hat{L}_j - \ell_j^*| \geq \alpha_{\log}/(8n)] \leq \beta/(2n). \quad (31)$$

Union-bounding over $j \in [n]$, the event

$$\mathcal{E}_L = \{|\hat{L}_j - \ell_j^*| \leq \alpha_{\log}/(8n) \text{ for all } j\} \quad (32)$$

holds with probability $\geq 1 - \beta/2$.

On $\mathcal{E}_\tau \cap \mathcal{E}_L$, the triangle inequality gives, for every j, u ,

$$|C[j, u] - C^*[j, u]| \leq |\hat{L}_j - \ell_j^*| + |\hat{\tau}_j(u) - \tau_{i_j,j,v_j}(u)| \leq \alpha_{\log}/(4n). \quad (33)$$

Hence for every permutation $\sigma \in S_n$,

$$\left| \sum_j C[j, \sigma(j)] - \sum_j C^*[j, \sigma(j)] \right| \leq n \cdot \alpha_{\log}/(4n) = \alpha_{\log}/4. \quad (34)$$

For any $\sigma \neq \varphi$, by Theorem 3.5,

$$\sum_j C^*[j, \sigma(j)] - \sum_j C^*[j, \varphi(j)] \geq \alpha_{\log}. \quad (35)$$

Thus,

$$\sum_j C[j, \sigma(j)] - \sum_j C[j, \varphi(j)] \geq \alpha_{\log} - 2 \cdot \alpha_{\log}/4 = \alpha_{\log}/2 > 0. \quad (36)$$

The Hungarian algorithm returns the unique-cost-minimizing assignment, which is φ . The verification step in Algorithm 1 accepts since $\varphi \in \mathcal{R}(x)$.

Let $\mathcal{E}_{\text{leak}}$ be the event that $(x, \varphi) \sim \mathcal{I}$ satisfies Theorem 3.5, and $\mathcal{E}_{\text{find}}$ be the event that FINDCOORDS outputs the correct advice. By hypothesis $\Pr[\mathcal{E}_{\text{leak}}] \geq \mu$, and conditional on $\mathcal{E}_{\text{leak}}$, Theorem B.2 gives $\Pr[\mathcal{E}_{\text{find}} \mid \mathcal{E}_{\text{leak}}] \geq q$. Steps 1–4 above show that conditional on $\mathcal{E}_{\text{leak}} \cap \mathcal{E}_{\text{find}}$, Algorithm 1 returns $\hat{\varphi} = \varphi$ with probability $\geq 1 - \beta = 3/4$. Hence

$$\Pr_{\mathcal{I}, P}[\text{EXTRACT}(x, \psi_1, \dots, \psi_N) \in \mathcal{R}(x)] \geq \mu \cdot q \cdot 3/4 > \text{negl}(n). \quad (37)$$

This is the hypothesis of Theorem 3.4. By that theorem, P is not zero-knowledge.

Resource accounting. The total sample complexity is $N = \text{poly}(n)$ oracle queries for table estimation. The Hungarian algorithm is also polynomial in n ($O(n^3)$). $\square$

C Spectral Deviation from Honest Behavior: Proofs

This appendix proves Theorem 3.7 and Theorem 3.8 from the main text. The argument proceeds in four steps:

1. We characterize the honest GMW-GI prover’s log-probability matrix as the constant rank-one matrix (Theorem C.1).
2. We state the empirical assumption of normalized power-law spectral decay (Theorem C.2), drawn from Golowich et al. [11].
3. We derive two consequences of this assumption: polynomial ϵ -rank and a non-trivial second singular value (Theorem C.3).
4. We combine these with a Frobenius-norm lower bound for probability distributions (Theorem C.4) and a spectral-norm scaling argument (Theorem C.5) to prove the Frobenius deviation theorem (Theorem C.7) and the average χ^2 corollary (Theorem C.8).

C.1 Setup and notation

We work on the $c = 0$ challenge branch of GMW-GI, where the prover’s response is the commitment permutation $\psi \in S_n$. Recall that an honest prover samples $\psi \sim \text{Unif}(S_n)$ independently of (x, φ) , while trained provers may produce any distribution over S_n .

We study the log-probability matrix

$$X_M[\psi, (x, \varphi)] = \log P[\psi \mid x, \varphi] \quad (38)$$

indexed by $\psi \in S_n$ (rows) and $(x, \varphi) \in \mathcal{V}$ (columns), where $\mathcal{V} = \{(x, \varphi) : \varphi \in \mathcal{R}(x)\}$ is the set of valid instance-witness pairs. We assume a subset of d_1 rows and d_2 columns. Write $p_{x, \varphi}(\psi) = P[\psi \mid x, \varphi]$ for the prover’s response distribution at (x, φ) , and X_M^{honest} for the log-probability matrix of the honest prover.

C.2 The honest prover

Proposition C.1 (Honest GMW-GI prover gives a rank-one matrix). *For an honest GMW-GI prover that samples $\psi \sim \text{Unif}(S_n)$ independently of (x, φ) , the log-probability matrix X_M^{honest} has all entries equal to $-\log(n!)$. It has rank exactly one, with non-zero singular value $\sigma_1 = \log(n!) \cdot \sqrt{d_1 \cdot d_2}$ and $\sigma_k = 0$ for $k \geq 2$.*

Proof. By definition of honest sampling, $P^{\text{honest}}[\psi \mid x, \varphi] = 1/n!$ for every $\psi \in S_n$ and every $(x, \varphi) \in \mathcal{V}$. Therefore $X_M^{\text{honest}} = -\log(n!) \cdot \mathbf{1}_{d_1} \mathbf{1}_{d_2}^\top$, the outer product of all-ones vectors scaled by $-\log(n!)$. This rank-one matrix has a single non-zero singular value $\log(n!) \cdot \|\mathbf{1}_{d_1}\|_2 \cdot \|\mathbf{1}_{d_2}\|_2 = \log(n!) \cdot \sqrt{d_1 \cdot d_2}$. $\square$

C.3 Power-law spectral decay

Golowich et al. [11] measure the singular value spectrum of log-probability matrices for diverse modern transformer language models, including OLMo-1b and OLMo-7b. They observe that the *normalized* singular values follow a power-law decay

$$\sigma_k / \sigma_1 \approx k^{-\alpha} \quad (39)$$

for some model-dependent exponent $\alpha > 1/2$. This assumption is also used to develop frameworks in following papers to explain subliminal learning [1].

Assumption C.2 (Normalized power-law spectral decay). The trained GMW-GI prover’s log-probability matrix X_M has singular values whose ratio to the top singular value follows a power-law shape:

$$\tilde{c} \cdot k^{-\alpha} \leq \frac{\sigma_k(X_M)}{\sigma_1(X_M)} \leq \tilde{C} \cdot k^{-\alpha} \quad (40)$$

for all $1 \leq k \leq \min(d_1, d_2)$, where $\tilde{c}, \tilde{C} > 0$ are absolute constants (independent of n) with $\tilde{c} \leq 1 \leq \tilde{C}$, and $\alpha > 1/2$ is a constant exponent. The absolute scale $\sigma_1(X_M)$ may depend on n ; only the normalized shape is assumed constant.

C.4 Consequences of power-law decay

Lemma C.3 (Polynomial ϵ -rank and non-trivial σ_2/σ_1). *Under Theorem C.2 with parameters $\tilde{c}, \tilde{C}, \alpha$:*

- (i) (Polynomial ϵ -rank.) *For any $\tilde{\epsilon} \geq 1/\text{poly}(n)$,*

$$\sum_{k > d(\tilde{\epsilon})} \sigma_k(X_M)^2 \leq \tilde{\epsilon}^2 \cdot \|X_M\|_F^2, \quad (41)$$

where $d(\tilde{\epsilon}) \leq C_\alpha \cdot \tilde{\epsilon}^{-2/(2\alpha-1)}$ and $C_\alpha = ((\tilde{C}/\tilde{c})^2/(2\alpha-1))^{1/(2\alpha-1)}$ is independent of n . In particular, $d(\tilde{\epsilon}) \leq \text{poly}(1/\tilde{\epsilon})$.

(ii) (Non-trivial second component.)

$$\frac{\sigma_2(X_M)}{\sigma_1(X_M)} \geq \tilde{c} \cdot 2^{-\alpha}, \quad (42)$$

a positive constant depending only on $\tilde{c}, \alpha$.

Proof. (i) By the upper bound in Theorem C.2, $\sigma_k(X_M)^2 \leq \tilde{C}^2 k^{-2\alpha} \sigma_1(X_M)^2$. The tail sum is bounded by integral comparison (valid since $2\alpha > 1$):

$$\sum_{k>d} \sigma_k(X_M)^2 \leq \tilde{C}^2 \sigma_1(X_M)^2 \sum_{k>d} k^{-2\alpha} \leq \tilde{C}^2 \sigma_1(X_M)^2 \cdot \frac{d^{1-2\alpha}}{2\alpha-1}. \quad (43)$$

By the lower bound in Theorem C.2, $\sigma_k(X_M)^2 \geq \tilde{c}^2 k^{-2\alpha} \sigma_1(X_M)^2$, so

$$\|X_M\|_F^2 = \sum_k \sigma_k(X_M)^2 \geq \tilde{c}^2 \sigma_1(X_M)^2, \quad (44)$$

using $\sum_{k \geq 1} k^{-2\alpha} \geq 1$. Taking the ratio,

$$\frac{\sum_{k>d} \sigma_k(X_M)^2}{\|X_M\|_F^2} \leq \frac{(\tilde{C}/\tilde{c})^2 \cdot d^{1-2\alpha}}{2\alpha-1}. \quad (45)$$

The factor $\sigma_1(X_M)^2$ cancels: the ratio depends only on $\tilde{C}/\tilde{c}$ and α . For this to be at most $\tilde{\epsilon}^2$,

$$d \geq \left(\frac{(\tilde{C}/\tilde{c})^2}{(2\alpha-1)\tilde{\epsilon}^2} \right)^{1/(2\alpha-1)} = C_\alpha \cdot \tilde{\epsilon}^{-2/(2\alpha-1)}. \quad (46)$$

Setting $d(\tilde{\epsilon}) = \lceil C_\alpha \tilde{\epsilon}^{-2/(2\alpha-1)} \rceil$ gives the bound.

(ii) By the lower bound in Theorem C.2 applied at $k=2$, $\sigma_2(X_M) \geq \tilde{c} \cdot 2^{-\alpha} \cdot \sigma_1(X_M)$. Dividing by $\sigma_1(X_M)$ gives the stated ratio. $\square$

C.5 Spectral norm scaling

We now establish a polynomial lower bound on $\sigma_1(X_M)$ in absolute terms, derived from the natural spectral scale of probability distributions on S_n combined with the normalized power-law shape.

Lemma C.4 (Frobenius lower bound for probability distributions). *For any prover producing valid probability distributions $\{p_{x,\varphi}\}$ over S_n ,*

$$\|X_M\|_F^2 \geq d_1 \cdot d_2 \cdot (\log n!)^2. \quad (47)$$

Proof. Fix a column $(x, \varphi) \in \mathcal{V}$. By the AM-GM inequality applied to $\{p_{x,\varphi}(\psi) : \psi \in S_n\}$,

$$\frac{1}{d_1} \sum_{d_1} p_{x,\varphi}(\psi) \geq \left(\prod_{d_1} p_{x,\varphi}(\psi) \right)^{1/d_1}. \quad (48)$$

Since $\sum_{d_1} p_{x,\varphi}(\psi) = \frac{d_1}{n!}$, we have $\prod_{d_1} p_{x,\varphi}(\psi) \leq (1/n!)^{d_1}$. Taking logarithms,

$$\sum_{d_1} \log p_{x,\varphi}(\psi) \leq -d_1 \log(n!), \quad (49)$$

so $|\sum_{d_1} \log p_{x,\varphi}(\psi)| \geq d_1 \log(n!)$. By Cauchy-Schwarz applied with the all-ones vector,

$$\sum_{d_1} (\log p_{x,\varphi}(\psi))^2 \geq \frac{(\sum_{d_1} \log p_{x,\varphi}(\psi))^2}{d_1} \geq d_1 (\log n!)^2. \quad (50)$$

Summing over all d_2 columns gives $\|X_M\|_F^2 \geq d_1 \cdot d_2 \cdot (\log n!)^2$. $\square$

Lemma C.5 (Spectral norm scaling). *Under Theorem C.2,*

$$\sigma_1(X_M)^2 \geq \frac{\|X_M\|_F^2}{\tilde{C}^2 \cdot \zeta(2\alpha)} \geq \frac{d_1 \cdot d_2 \cdot (\log n!)^2}{\tilde{C}^2 \cdot \zeta(2\alpha)}, \quad (51)$$

where $\zeta(s) = \sum_{k \geq 1} k^{-s}$ is the Riemann zeta function (finite for $s = 2\alpha > 1$).

Proof. By the upper bound in Theorem C.2,

$$\|X_M\|_F^2 = \sum_k \sigma_k(X_M)^2 \leq \tilde{C}^2 \sigma_1(X_M)^2 \sum_k k^{-2\alpha} \leq \tilde{C}^2 \sigma_1(X_M)^2 \cdot \zeta(2\alpha). \quad (52)$$

Rearranging gives $\sigma_1(X_M)^2 \geq \|X_M\|_F^2 / (\tilde{C}^2 \zeta(2\alpha))$. Combining with Theorem C.4 gives the second inequality. $\square$

C.6 Bounded probability assumption

For the χ^2 corollary, we require an additional regularity assumption on the trained prover's distributions. (We re-state it here in the spectral context; the same name is used for the related assumption in Section B with a different normalization.)

Assumption C.6 (Bounded probability for spectral analysis). For every $(x, \varphi) \in \mathcal{V}$ and every $\psi \in S_n$,

$$p_{x,\varphi}(\psi) \geq 1/(2n!). \quad (53)$$

C.7 Frobenius spectral deviation from honest behavior

Theorem C.7 (Frobenius spectral deviation; formal version of Theorem 3.7). *Under Theorem C.2, the trained prover's log-probability matrix differs from the honest baseline by*

$$\|X_M - X_M^{\text{honest}}\|_F \geq \sigma_2(X_M) \geq \tilde{c} \cdot 2^{-\alpha} \cdot \sigma_1(X_M) \geq \Omega\left(\log(n!) \cdot \sqrt{d_1 \cdot d_2}\right), \quad (54)$$

where the implicit constant in $\Omega(\cdot)$ depends only on $\tilde{c}, \tilde{C}, \alpha$. The deviation matrix $\Delta = X_M - X_M^{\text{honest}}$ has rank at least $\text{rank}(X_M) - 1 \geq 1$.

Proof. By Theorem C.1, X_M^{honest} has rank one. By the Eckart–Young theorem, the best rank-one approximation $X^{(1)}$ of X_M minimizes Frobenius distance among all rank-one matrices, so for any rank-one matrix Y , $\|X_M - Y\|_F \geq \|X_M - X^{(1)}\|_F$. Applied with $Y = X_M^{\text{honest}}$,

$$\|\Delta\|_F \geq \|X_M - X^{(1)}\|_F = \sqrt{\sum_{k \geq 2} \sigma_k(X_M)^2} \geq \sigma_2(X_M). \quad (55)$$

By Theorem C.3(ii), $\sigma_2(X_M) \geq \tilde{c} \cdot 2^{-\alpha} \cdot \sigma_1(X_M)$. By Theorem C.5, $\sigma_1(X_M) \geq \log(n!) \sqrt{d_1 d_2} / (\tilde{C} \sqrt{\zeta(2\alpha)})$. Combining,

$$\|\Delta\|_F \geq \frac{\tilde{c} \cdot 2^{-\alpha}}{\tilde{C} \sqrt{\zeta(2\alpha)}} \cdot \log(n!) \cdot \sqrt{d_1 d_2}, \quad (56)$$

where the leading coefficient depends only on $\tilde{c}, \tilde{C}, \alpha$.

For the rank statement, by subadditivity of rank, $\text{rank}(\Delta) \geq \text{rank}(X_M) - \text{rank}(X_M^{\text{honest}}) = \text{rank}(X_M) - 1$. Since Theorem C.3(ii) implies $\sigma_2(X_M) > 0$, $\text{rank}(X_M) \geq 2$, so $\text{rank}(\Delta) \geq 1$. $\square$

C.8 Average χ^2 deviation from uniform

Corollary C.8 (Formal version of Theorem 3.8). *Under Theorems C.2 and C.6, the trained prover's response distribution is far from uniform on average with χ^2 divergence:*

$$\mathbb{E}_{(x,\varphi) \sim \text{Unif}(\mathcal{V})} [\chi^2(p_{x,\varphi}, \text{Unif}(S_n))] \geq \Omega((\log n!)^2) = \text{poly}(n), \quad (57)$$

where the implicit constant depends only on $\tilde{c}, \tilde{C}, \alpha$.

Proof. The Frobenius deviation decomposes column-wise:

$$\|X_M - X_M^{\text{honest}}\|_F^2 = \sum_{(x,\varphi) \in d_2} \sum_{\psi \in d_1} (\log p_{x,\varphi}(\psi) - \log(1/n!))^2. \quad (58)$$

By Theorem C.7, $\|X_M - X_M^{\text{honest}}\|_F^2 \geq (\tilde{c} \cdot 2^{-\alpha})^2 \sigma_1(X_M)^2$. Averaging over $(x, \varphi) \in d_2$ and assuming this is uniformly random over $\mathcal{V}$,

$$\mathbb{E}_{(x,\varphi)} \left[\sum_{\psi \in d_1} (\log p_{x,\varphi}(\psi) - \log(1/n!))^2 \right] \geq \frac{(\tilde{c} \cdot 2^{-\alpha})^2 \sigma_1(X_M)^2}{d_2}. \quad (59)$$

By the mean value theorem applied to $z \mapsto e^z$, for any $a, b \in \mathbb{R}$, $|e^a - e^b| \geq \min(e^a, e^b) \cdot |a - b|$. Setting $a = \log p_{x,\varphi}(\psi)$ and $b = \log(1/n!)$,

$$|p_{x,\varphi}(\psi) - 1/n!| \geq \min(p_{x,\varphi}(\psi), 1/n!) \cdot |\log p_{x,\varphi}(\psi) - \log(1/n!)|. \quad (60)$$

By Theorem C.6, $\min(p_{x,\varphi}(\psi), 1/n!) \geq 1/(2n!)$. Squaring,

$$(p_{x,\varphi}(\psi) - 1/n!)^2 \geq \frac{1}{(2n!)^2} (\log p_{x,\varphi}(\psi) - \log(1/n!))^2. \quad (61)$$

$$\begin{aligned}
\chi^2(p_{x,\varphi}, \text{Unif}(S_n)) &= n! \sum_{\psi \in d_1} (p_{x,\varphi}(\psi) - 1/n!)^2, \quad \text{which only makes sense for } d_1 = |S_n| \\
&\geq \frac{n!}{(2n!)^2} \sum_{\psi \in d_1} (\log p_{x,\varphi}(\psi) - \log(1/n!))^2 \\
&= \frac{1}{4n!} \sum_{\psi \in d_1} (\log p_{x,\varphi}(\psi) - \log(1/n!))^2.
\end{aligned}$$

Taking expectation over $(x, \varphi) \sim \text{Unif}(\mathcal{V})$,

$$\mathbb{E}[\chi^2] \geq \frac{1}{4n!} \cdot \frac{(\tilde{c} \cdot 2^{-\alpha})^2 \sigma_1(X_M)^2}{d_2}. \quad (62)$$

By Theorem C.5, $\sigma_1(X_M)^2 \geq d_1 \cdot d_2 \cdot (\log n!)^2 / (\tilde{C}^2 \zeta(2\alpha))$ and we said the probability distribution only makes sense if $d_1 = |S_n|$. Substituting (with $|S_n| = n!$),

$$\mathbb{E}[\chi^2] \geq \frac{1}{4n!} \cdot \frac{(\tilde{c} \cdot 2^{-\alpha})^2}{d_2} \cdot \frac{n! \cdot d_2 \cdot (\log n!)^2}{\tilde{C}^2 \zeta(2\alpha)} = \frac{(\tilde{c} \cdot 2^{-\alpha})^2 \cdot (\log n!)^2}{4\tilde{C}^2 \zeta(2\alpha)}. \quad (63)$$

The remaining expression has $(\log n!)^2 = \Theta((n \log n)^2) = \text{poly}(n)$, multiplied by absolute constants depending only on $\tilde{c}, \tilde{C}, \alpha$. $\square$

Theorem C.8 establishes that the trained prover’s response distribution is far from uniform: in χ^2 divergence, the average deviation across (x, φ) pairs is polynomially large in n . We emphasize that $\chi^2 \geq \text{poly}(n)$ does not directly translate to polynomial-sample identity testing for the full distribution: distinguishing $p_{x,\varphi}$ from $\text{Unif}(S_n)$ via samples requires $\Omega(\sqrt{|S_n|}/\chi^2) = \Omega(\sqrt{n!}/\text{poly}(n))$ samples, which is exponential in n . The χ^2 bound is a structural divergence statement characterizing the trained prover as non-uniform, not an operational sample complexity bound. Polynomial-time distinguishability is the empirical content captured by Theorem 3.5 in the main paper, which is verified at $n \leq 9$ in Section 5.

D Implementation Details

This section of the appendix describes the model architecture, training data and objective, simplifications made for tractability, and the sample budgets used by the extractor.

D.1 Architecture

We train a small decoder-only transformer (TinyTransformer) with a hidden dimension of $d = 256$, $L = 8$ layers, $H = 4$ attention heads, and feedforward width of $4d = 1024$. The architecture utilizes pre-LayerNorm with GELU activations, a causal self-attention mask, and learned token and positional embeddings. The vocabulary is exactly the set of permutation values $\{0, 1, \dots, n-1\}$, which means there are no separator or padding, and the positional embedding table has size $4n$.

D.2 Sequence layout and training objective

Each training example consists of a length- $4n$ token sequence with four contiguous permutation blocks:

$$\left[\underbrace{\varphi_0, \dots, \varphi_{n-1}}_n \mid \underbrace{\psi_0, \dots, \psi_{n-1}}_n \mid \underbrace{\psi_0^{-1}, \dots, \psi_{n-1}^{-1}}_n \mid \underbrace{(\varphi \circ \psi^{-1})_0, \dots, (\varphi \circ \psi^{-1})_{n-1}}_n \right]. \quad (64)$$

We trained the model as a causal language model with cross-entropy loss applied to the latter three blocks (ψ , ψ^{-1} , and $\varphi \circ \psi^{-1}$). The witness block φ is input-only, so no loss is applied at those positions. The total training loss is then calculated as the sum of the three per-block cross-entropies. It’s worth noting that the auxiliary blocks ψ^{-1} and $\varphi \circ \psi^{-1}$ are deterministic functions of (φ, ψ) . We included them for the purposes of the same prover being able to serve responses on the $c = 1$ branch, which reveals $\sigma = \psi \circ \varphi^{-1}$ and requires both the φ and ψ values to perform the computation.

D.3 Conditioning simplification: omitting (G_0, G_1)

In the complete GMW-GI protocol, the prover sees the instance (G_0, G_1) in addition to φ . In our setting, We omit the graphs and instead condition only on φ , such that the prover learns $\varphi \mapsto P[\psi \mid \varphi]$ rather than $(x, \varphi) \mapsto P[\psi \mid x, \varphi]$. This shortens the input sequence from $\Theta(n^2)$ to $4n$ and avoids representation of graph adjacency in τ . The structural property exploited by our attacks, that $\psi \in S_n$ is exposed verbatim on the $c = 0$ branch, is preserved.

We empirically verified at $n = 4$ and $n = 5$ that this simplification does not change the witness-recovery rates of our extractors (Table 3).

Table 3: Top- n recovery rate (%) of the best single-witness extractor with and without (G_0, G_1) conditioning. Differences are within sampling noise.

Conditioning	$n = 4$	$n = 5$
With (G_0, G_1) (full GMW-GI input)	19.8	7.05
Without (witness-only, used in this paper)	20.8	5.0

D.4 Per- n training configuration

We train one prover per value of n . As n grows, we increase both the size of the training set (such that the model has sufficient pairs of (φ, ψ) to generalize through the auxiliary tasks) as well as the number of gradient updates. Table 4 reports the configuration. All of the other hyperparameters are shared across n and defined as follows: AdamW optimizer with a learning rate of 3×10^{-4} , no weight decay, and no learning rate schedule. Finally, the models are trained on a single NVIDIA RTX 3090.

Table 4: Per- n training configuration. Dataset size is the number of (φ, ψ) training pairs (sampled iid uniform from $S_n \times S_n$); iterations is the number of gradient updates at batch size 32.

	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$
Dataset size	100	100	200	500	10000	20000
Steps	2500	2500	5000	10000	50000	100000
Batch Size	8	8	8	8	64	64

D.5 Output validity diagnostics

For each trained prover, we report teacher-forced diagnostics that confirm that our model has internalized the structural relations between the four blocks (Table 5):

- ψ *valid*: the fraction of unconstrained autoregressive samples at which the emitted ψ is a valid permutation of $[n]$.
- ψ^{-1} *correct* | ψ : with (φ, ψ) teacher-forced, the fraction of test instances at which the prediction of $\arg\max$ in every position ψ^{-1} matches $\text{argsort}(\psi)$.

Table 5: Per-instance validity diagnostics for the baseline prover. Numbers are percent of test instances. Test set: 5000 random permutations for the ψ -validity diagnostic, 1000 for the conditional-correctness diagnostics.

Diagnostic	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$
ψ valid	99.78	97.76	94.80	97.54	99.96	99.98
ψ^{-1} correct ψ	98.00	91.00	99.40	100.00	100.00	100.00

D.6 Defense training

The simulator-aligned prover (Section 4.1) is trained with the same architecture, sequence layout, and hyperparameters as the baseline prover. However, for the simulator-aligned prover, the cross-entropy loss on the ψ block is replaced by KL divergence to the autoregressive factorization of $\text{Unif}(S_n)$: at position j within the ψ block, the target distribution is uniform over $[n] \setminus \{\psi_0, \dots, \psi_{j-1}\}$. The losses on the ψ^{-1} and $\varphi \circ \psi^{-1}$ blocks remain unchanged.

The witness-masked prover (Section 4.2) utilizes the same training as the baseline. During inference, the attention mask is modified during generation of the ψ block, deterring those positions from attending to the φ block. We then revert to the standard causal mask for the ψ^{-1} and $\varphi \circ \psi^{-1}$ blocks.